

Introduction to Functional Medicine



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Robert Luby, MD disclosed no relevant financial relationships with any ineligible companies.

Monique T. Class, MS, ARNP-BC disclosed no relevant financial relationships with ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Recall the key features of functional medicine.

Clinical Focus

- What are the features of functional medicine, and how would you describe it?
- What is one doing when one is practicing functional medicine?



Functional Medicine

What is it and what is its role?

- A comprehensive, systems-based approach to health and well-being.
- A catalyst in the transformation of healthcare.
- A vital partner to conventional medicine.



Functional Medicine

What type of medical practice is it?

- Addresses the root causes of disease and restores healthy function through a personalized patient experience.
- Understands each patient's genetic, environmental, and lifestyle influences to promote optimal health and well-being.
- Does not replace but is synergistic with and supportive of conventional medical practice models.



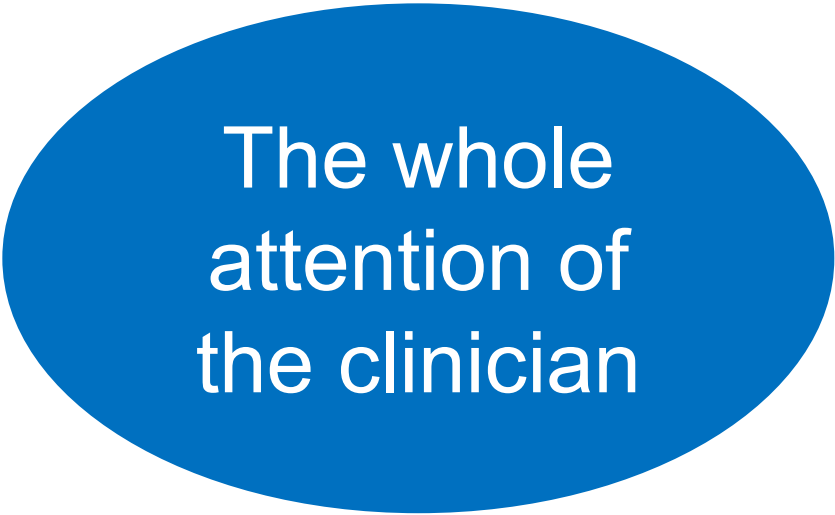
Functional Medicine

What is the clinical approach?

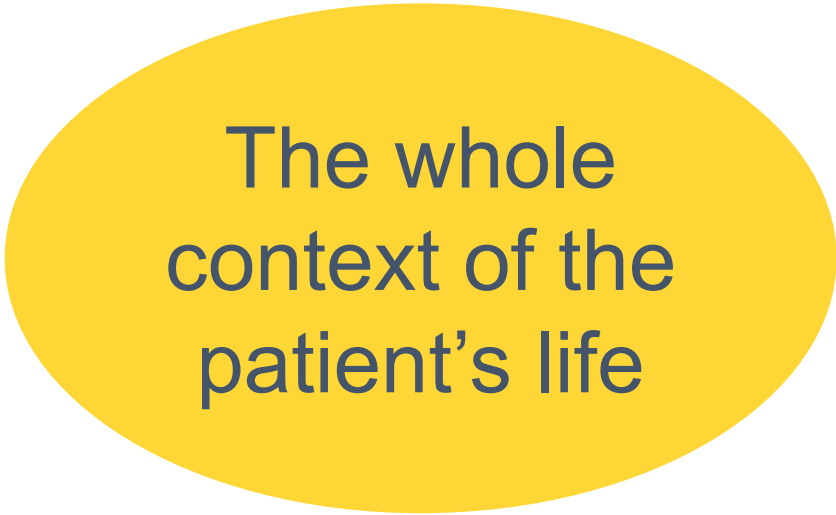
- An evaluative method that identifies the root causes of symptoms, diseases, and physiological dysfunctions.
- A systems-based therapeutic approach to address those symptoms, diseases, and dysfunctions.
- Based on science, evidence, systems biology, clinical reasoning, patient uniqueness, and patient preferences.



Functional Medicine Involves the Whole Person



The whole
attention of
the clinician



The whole
context of the
patient's life

Features of Functional Medicine

What is one doing when one is practicing functional medicine?

- Gathering oneself prior to each visit to fully attend to the patient.
- Gathering and understanding the patient's current circumstances and context.
- Gathering an understanding of what matters to the patient; their goals, values, priorities, and preferences.



Functional Medicine Evaluates the Patient Beyond the “Review of Systems”

What are the
symptoms?



What are the
key health
factors of the
patient's life?

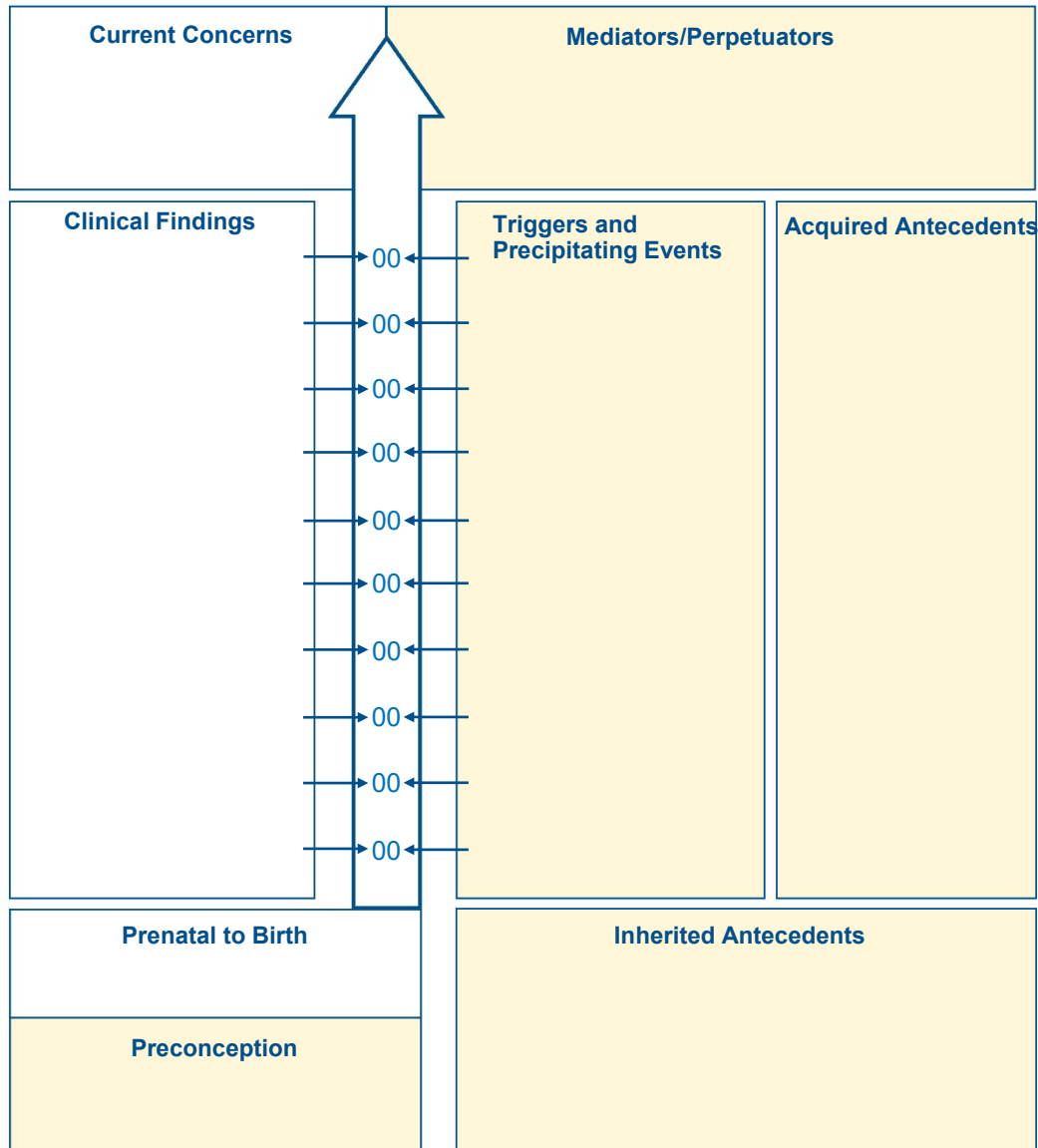
Features of Functional Medicine

What is one doing when one is practicing functional medicine?

Gathering all the factors in a patient's life that constitute the interplay of how genetics, lifestyle, and environment have contributed to their current concerns.

- Antecedents (predispositions)
- Triggers (activators)
- Mediators (perpetuators)
- Mental, emotional, and spiritual factors

Functional Medicine Timeline

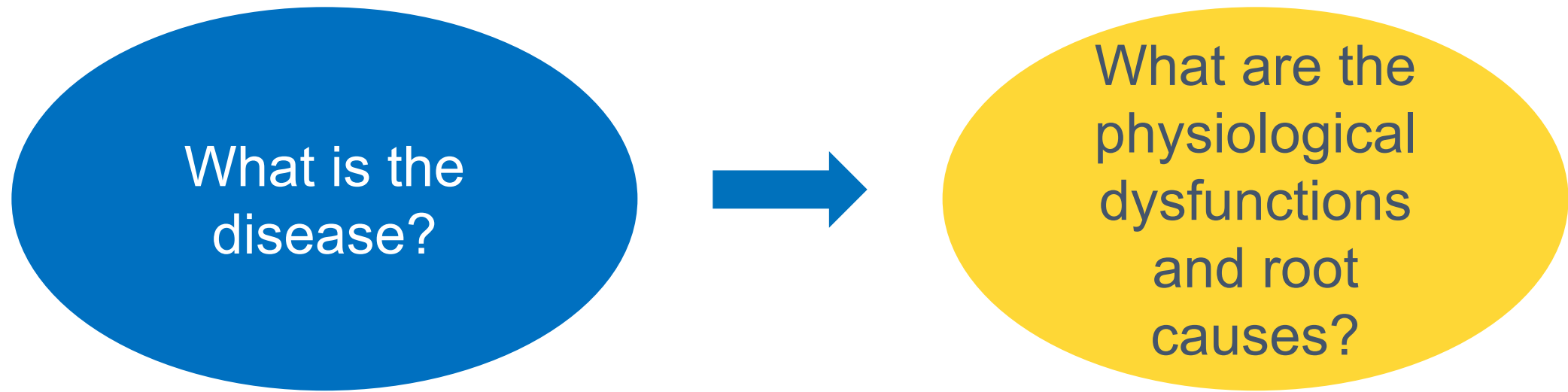


**The clinical tool to
capture and
document the key
health events and
root cause factors
in a patient's life.**

Long Description: Functional Medicine Timeline

- Available toolkit item – Functional Medicine Timeline

Functional Medicine Diagnoses the Patient Beyond Disease



Features of Functional Medicine

What is one doing when one is practicing functional medicine?

- Organizing all of the factors of the patient's life timeline to identify patterns and understand the personalized root causes of physiological dysfunction across body systems.
- Generating an explanation of the patient's current concerns and clinical findings based on these root causes.

Functional Medicine Matrix

The point-of-care tool to organize the patient's clinical findings and identify patterns of dysfunction.

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

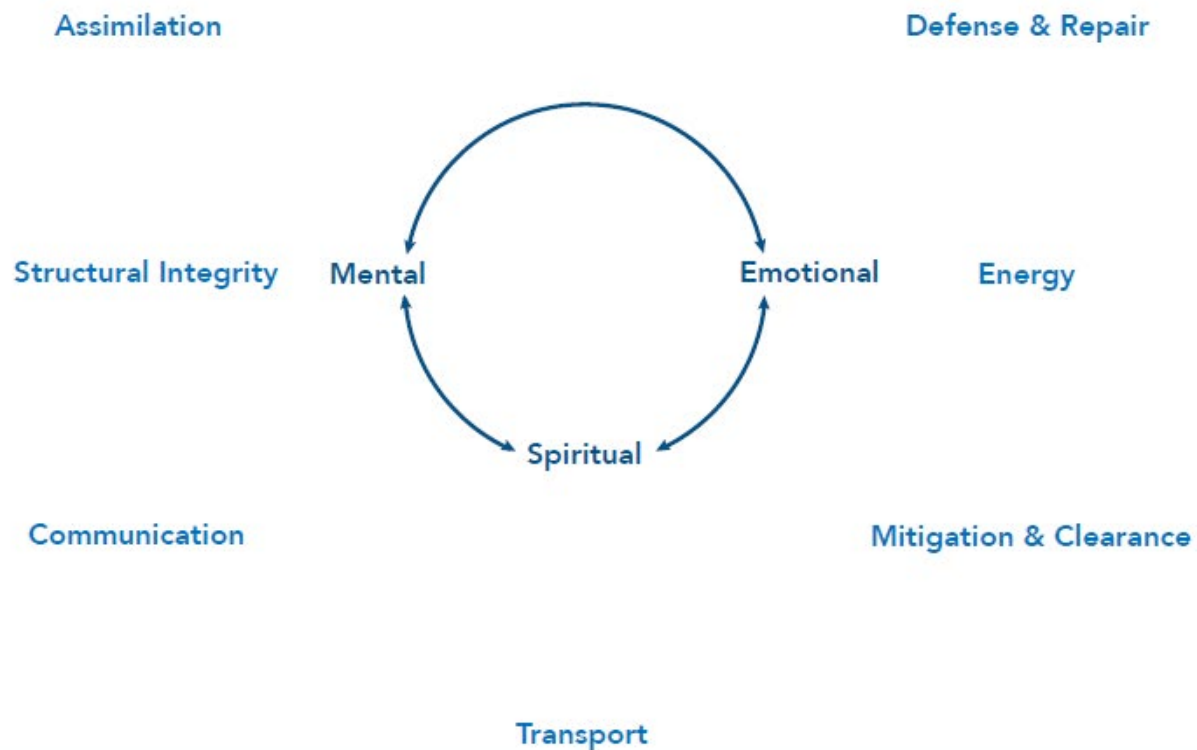
Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

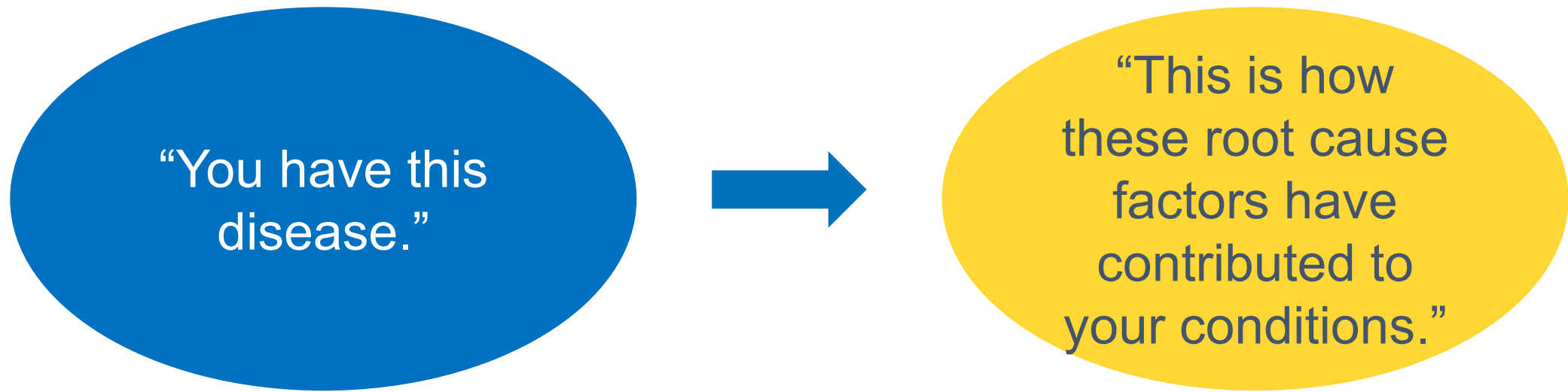


Long Description: Functional Medicine Matrix

- Available toolkit item – Functional Medicine Matrix

Functional Medicine Is Root Cause Medicine

Functional Medicine Is Personalized Medicine



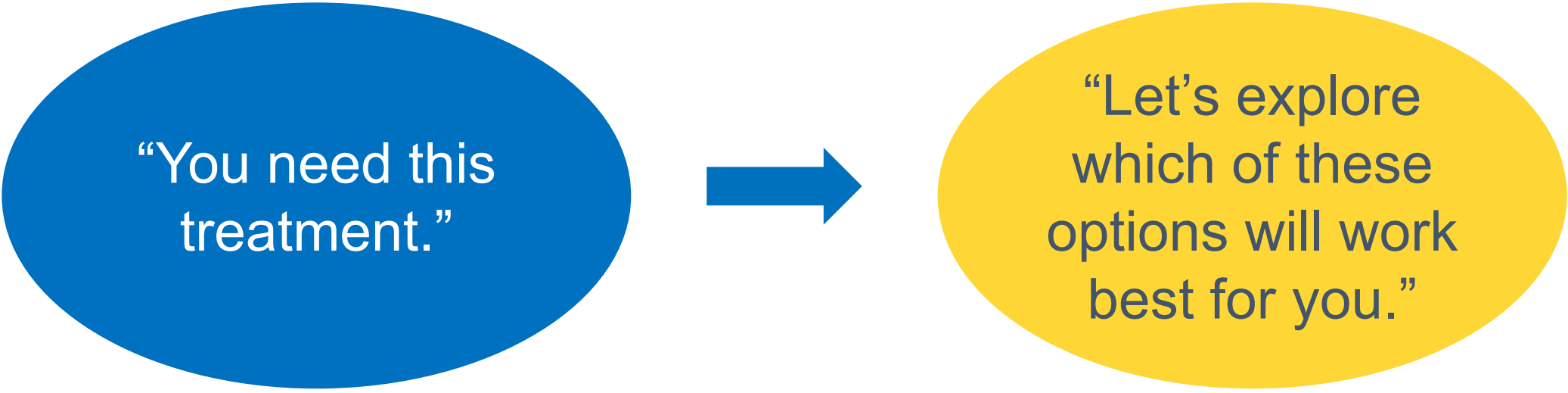
Features of Functional Medicine

What is one doing when one is practicing functional medicine?

Retelling the health story to the patient in a way that creates a shared understanding of the root causes of their conditions.



Functional Medicine Engenders a Therapeutic Partnership With the Patient



“You need this
treatment.”



“Let’s explore
which of these
options will work
best for you.”

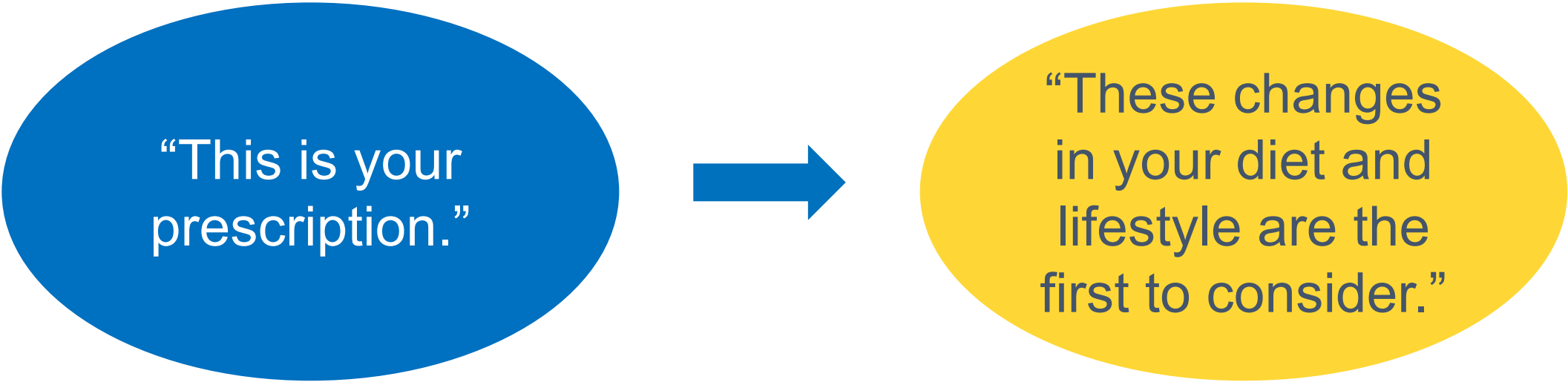
Features of Functional Medicine

What is one doing when one is practicing functional medicine?

- Identifying interventions that target the personalized root causes, physiological dysfunctions, and clinical imbalances of each patient.
- Collaborating in therapeutic partnership with the patient to co-prioritize interventions that would be the most feasible and effective for the patient based on evidence, clinical judgment, and patient preference.



Functional Medicine Leads With Safe, Valid, and Effective Interventions



“This is your
prescription.”



“These changes
in your diet and
lifestyle are the
first to consider.”

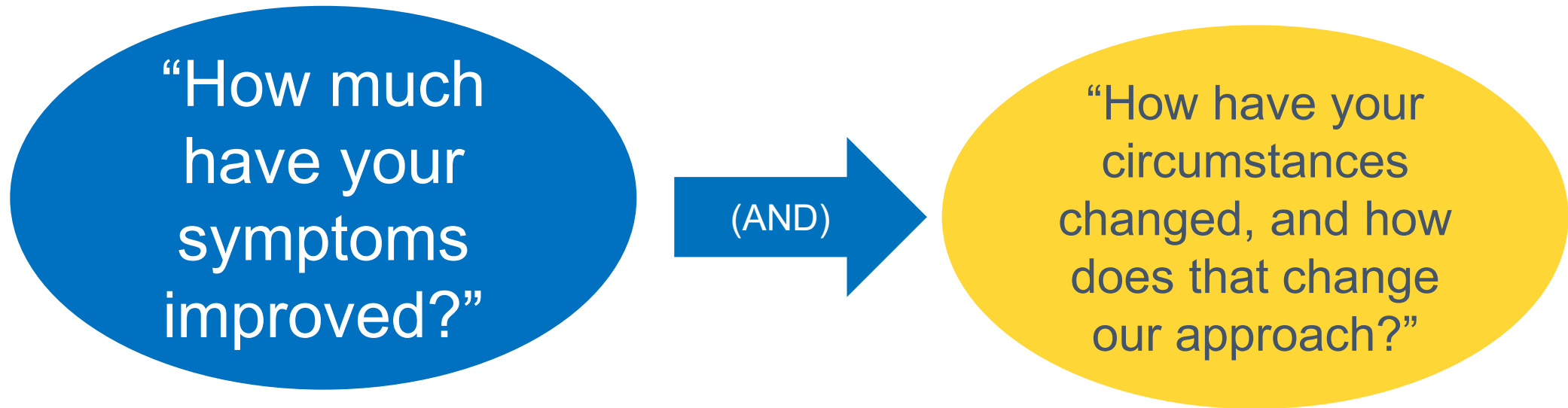
Features of Functional Medicine

What is one doing when one is practicing functional medicine?



Prioritizing diet and lifestyle changes as the foundational interventions that target the triggers, mediators, and root causes of the patient's conditions.

Functional Medicine Monitors the Whole Patient



Features of Functional Medicine

What is one doing when one is practicing functional medicine?

- Tracking the patient's response to interventions and changes in their circumstances that inform subsequent recommendations.



Functional Medicine Is Therapeutically Comprehensive

Dietary
Interventions

Lifestyle
Behavior
Changes

Medications

Supplements
and Integrative
Modalities

Functional Medicine Is Comprehensive

What is one doing when one is practicing functional medicine?

Keeping conventional guidelines in mind while layering in:

Evidence

Clinical Experience

Patient Preference



Functional Medicine Is Innovative and Personalized

Wearable
Devices

Shared
Medical
Appointments

Toolkit to
Educate,
Evaluate, and
Monitor

Interdisciplinary
Professional
Referrals

Functional Medicine Is Comprehensive

What is one doing when one is practicing functional medicine?

Utilizing myriad safe, valid, & effective personalized interventions based on:

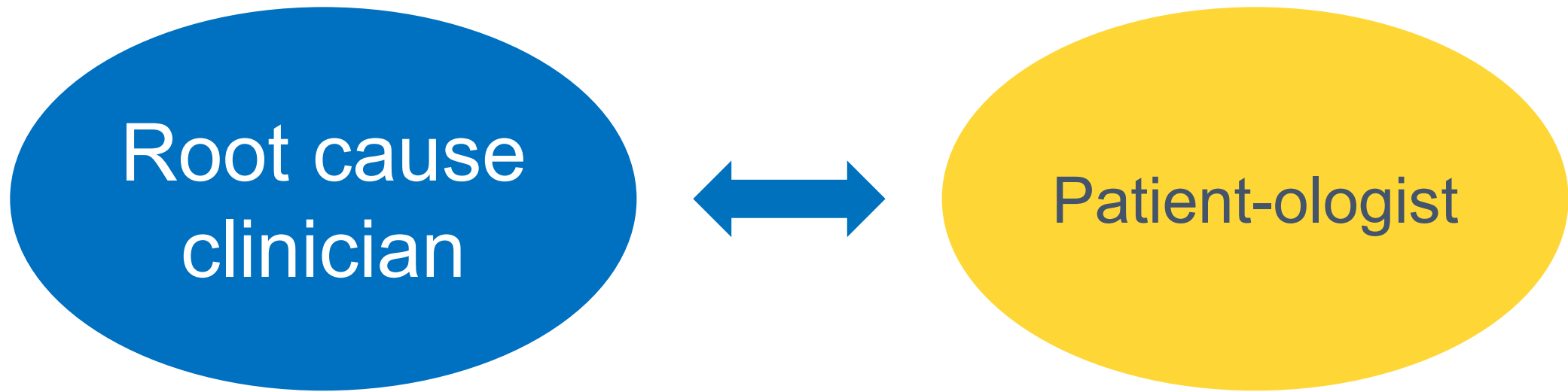
Evidence

Patient Root Causes

Patient Context



Functional Medicine Clinicians Specialize in Root Causes and the Patient



Clinical Focus

- What is functional medicine solving for?
- The role of functional medicine in addressing the global chronic disease epidemic.



Chronic Diseases in the US: Financial and Human Scope

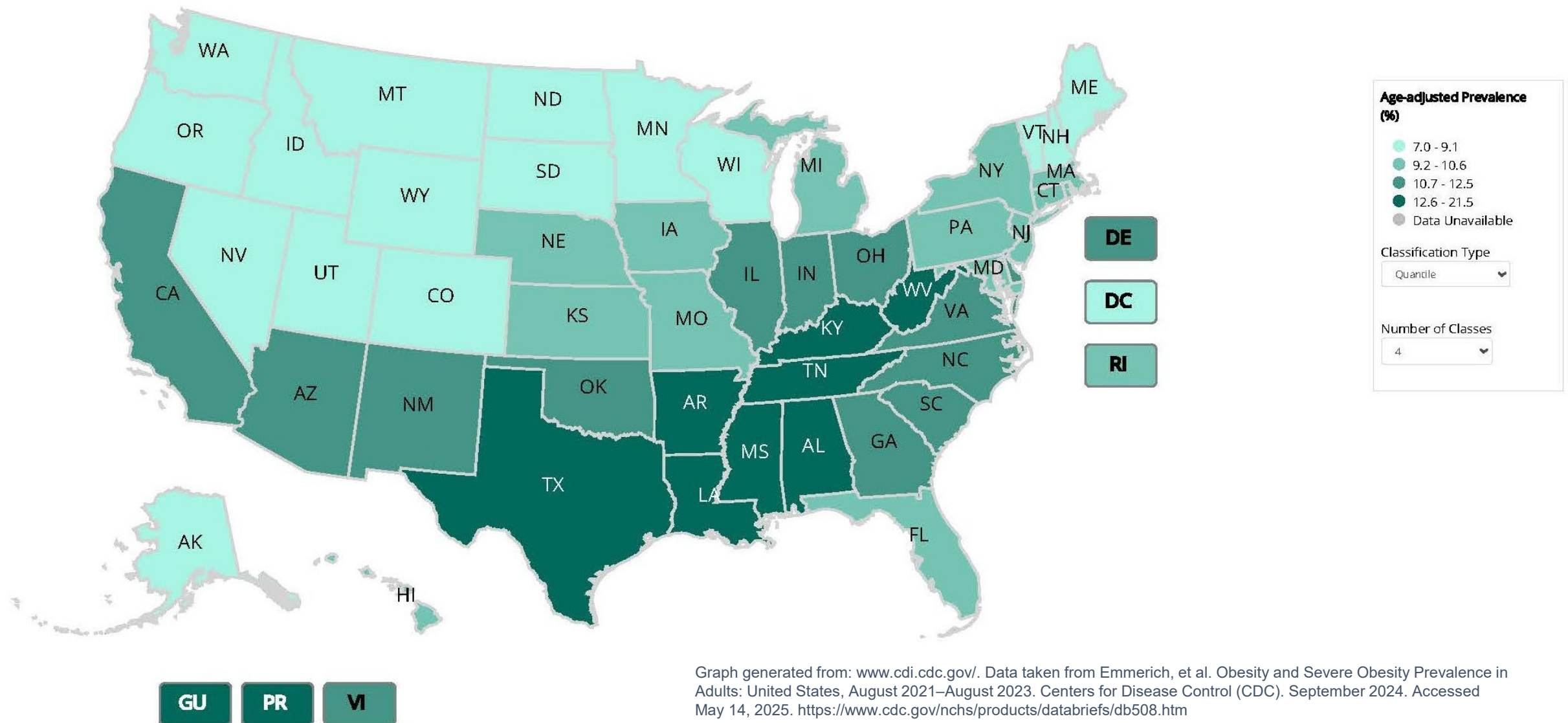
Of the annual \$4.1 trillion healthcare expenditure in the US, 90% is attributed to managing and treating chronic diseases and mental health conditions.

In the US:

- Five out of the ten leading causes of death are strongly associated with preventable and treatable chronic diseases.
- 129 million people have at least one major chronic health condition.
- 42% have two or more chronic health conditions.
- 12% have at least five chronic health conditions.



Age-Adjusted Prevalence of Diabetes in US Adults

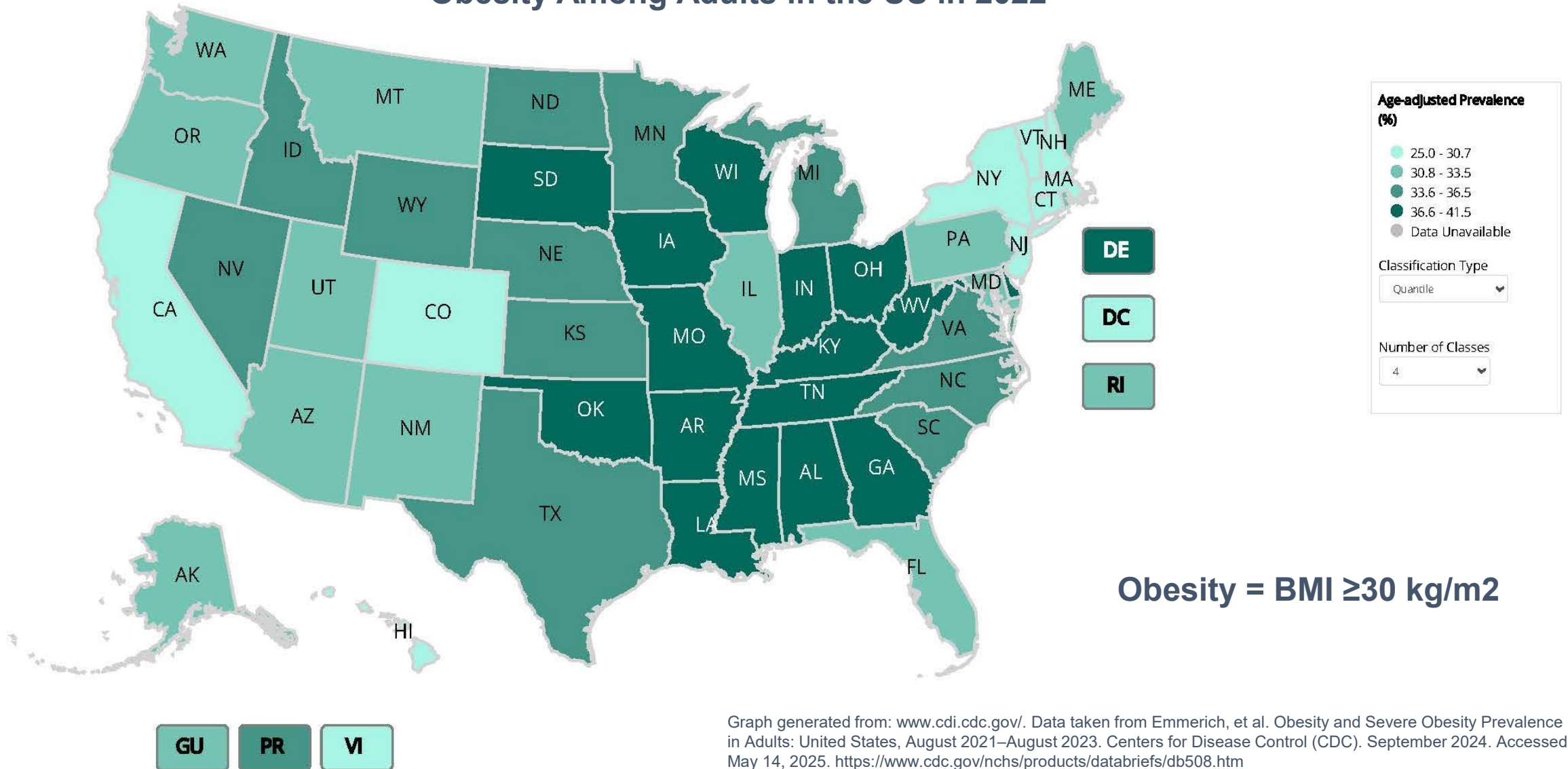


Long Description: Diabetes Prevalence

- This image depicts a map of the United States. Each state is color-coded based on age-adjusted prevalence of diabetes in U S adults. The states with a prevalence of 7-9.1% include: Washington, Oregon, Nevada, Idaho, Utah, Montana, Wyoming, Colorado, Alaska, North Dakota, South Dakota, Minnesota, Wisconsin, Washington D C, Vermont, New Hampshire, and Maine. The states with a prevalence of 9.2-10.6% include Nebraska, Kansas, Iowa, Missouri, Michigan, Pennsylvania, New York, Massachusetts, Connecticut, New Jersey, Rhode Island, Maryland, Florida, and Hawaii. The states and territories with a prevalence of 10.7-12.5% include: California, Arizona, New Mexico, Virgin Islands, Oklahoma, Illinois, Indiana, Ohio, Virginia, North Carolina, South Carolina, Georgia, and Delaware. The states and territories with a prevalence of 12.6-21.5% include Texas, Arkansas, Louisiana, Mississippi, Alabama, West Virginia, Kentucky, Tennessee, Puerto Rico, and Guam.

Age-Adjusted Prevalence of Obesity in US Adults

Obesity Among Adults in the US in 2022



Long Description: Obesity Prevalence

- This image depicts a map of the United States. Each state is color-coded based on age-adjusted prevalence of obesity in U S adults. Obesity is defined as a BMI greater than or equal to 30 kilograms per meter squared. States and territories with a prevalence of 25-30.7% include: California, Colorado, Hawaii, New York, Vermont, New Hampshire, Massachusetts, Connecticut, New Jersey, Washington D C, and the Virgin Islands. States with a prevalence of 30.8-33.5% include Washington, Oregon, Montana, Utah, Arizona, New Mexico, Alaska, Illinois, Maine, Pennsylvania, Rhode Island, Maryland, Florida, and Guam. States with a prevalence of 33.6-36.5% include Idaho, Nevada, Wyoming, North Dakota, Nebraska, Kansas, Texas, Minnesota, Michigan, Virginia, North Carolina, South Carolina, and Puerto Rico. States with a prevalence of 36.6-41.5% include South Dakota, Oklahoma, Wisconsin, Iowa, Missouri, Arkansas, Louisiana, Mississippi, Alabama, Georgia, Tennessee, Kentucky, Indiana, Ohio, West Virginia, and Delaware.

The US Versus 38 OECD Nations

Healthcare
costs per capita:
\$12,555 vs
<\$5,000

Greater
proportion of
GDP than any
other country

Lower life
expectancy
than OECD
average

The Chronic Disease Problem

It is projected that the cost of chronic disease will be **\$47 trillion globally by 2030.**

**\$47 trillion
globally
by 2030**



- Bloom DE, Cafiero ET, Jané-Llopis E, et al. *The Global Economic Burden of Non-communicable Diseases*. World Economic Forum; 2011. Accessed May 28, 2025. https://www3.weforum.org/docs/WEF_Harvard_HE_GlobalEconomicBurdenNonCommunicableDiseases_2011.pdf
- Bureau of Economic Analysis. Gross domestic product, third quarter 2024 (advance estimate). Published October 30, 2024. Accessed May 28, 2025. <https://www.bea.gov/news/2024/gross-domestic-product-third-quarter-2024-advance-estimate>
- Hacker K. The burden of chronic disease [published correction appears in *Mayo Clin Proc Innov Qual Outcomes*. 2024;9(1):100588. doi:10.1016/j.mayocpiqo.2024.11.005]. *Mayo Clin Proc Innov Qual Outcomes*. 2024;8(1):112-119. doi:10.1016/j.mayocpiqo.2023.08.005

Clinical Focus

- Research emerging to support the functional medicine model.
- Cost savings
- Patient-reported outcomes



Healthcare Costs

Value of the Functional Medicine Clinical Method

Cleveland Clinic

Compared FM shared medical appointments (SMAs) to FM individual appointments:

SMAs were \$508 more cost effective per patient over three months.

Ashland School District

Functional medicine vs conventional care for musculoskeletal complaints:

Cost savings of \$534 in pharmaceutical claims per patient over 11 years.

NCCIH

Computer modeling for a hypothetical patient from age 40 to 80:

Whole person/functional medicine care would result in >\$300,000 healthcare cost savings.

- Beidelschies M, Alejandro-Rodriguez M, Guo N, et al. Patient outcomes and costs associated with functional medicine-based care in a shared versus individual setting for patients with chronic conditions: a retrospective cohort study. *BMJ Open*. 2021;11(4):e048294. doi:10.1136/bmjopen-2020-048294
- D'Adamo CR, Grebinsky K, Roedersheimer G, et al. Functional medicine care was associated with lower pharmacy claims costs among Ashland School District employee health plan participants with musculoskeletal disorders. *Altern Ther Health Med*. 2024;30(10):34-38.
- Herman PM, Pitcher MH, Langevin HM. Making a case for whole person health. *Glob Adv Integr Med Health*. 2024;13:27536130241293642. doi:10.1177/27536130241293642

Healthcare Outcomes

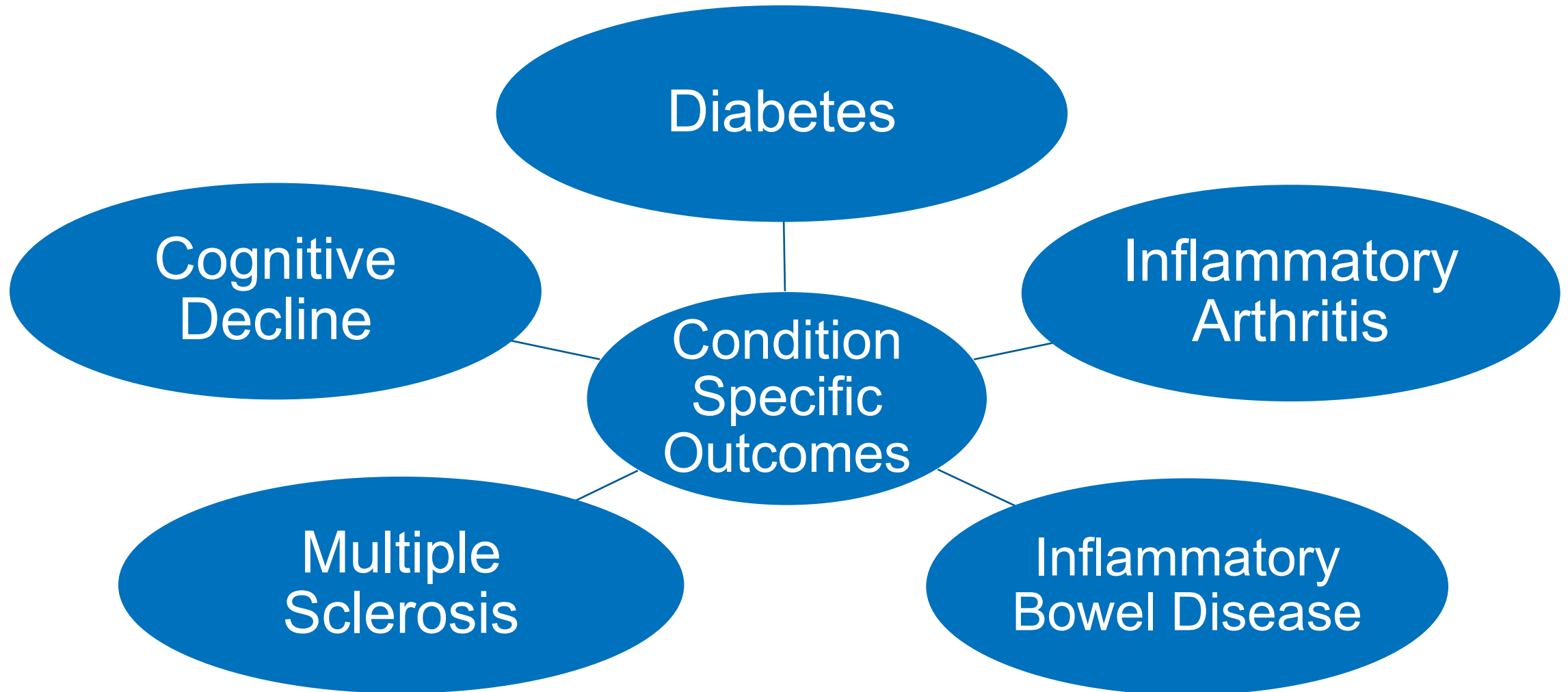
Value of the Functional Medicine Clinical Method

**Cleveland Clinic
12-month Clinical Study:
Functional medicine
model of care vs
primary care model**

Outcomes:
Significantly greater
improvements in patient-
reported outcomes
(PROMIS and GPH T-scores)

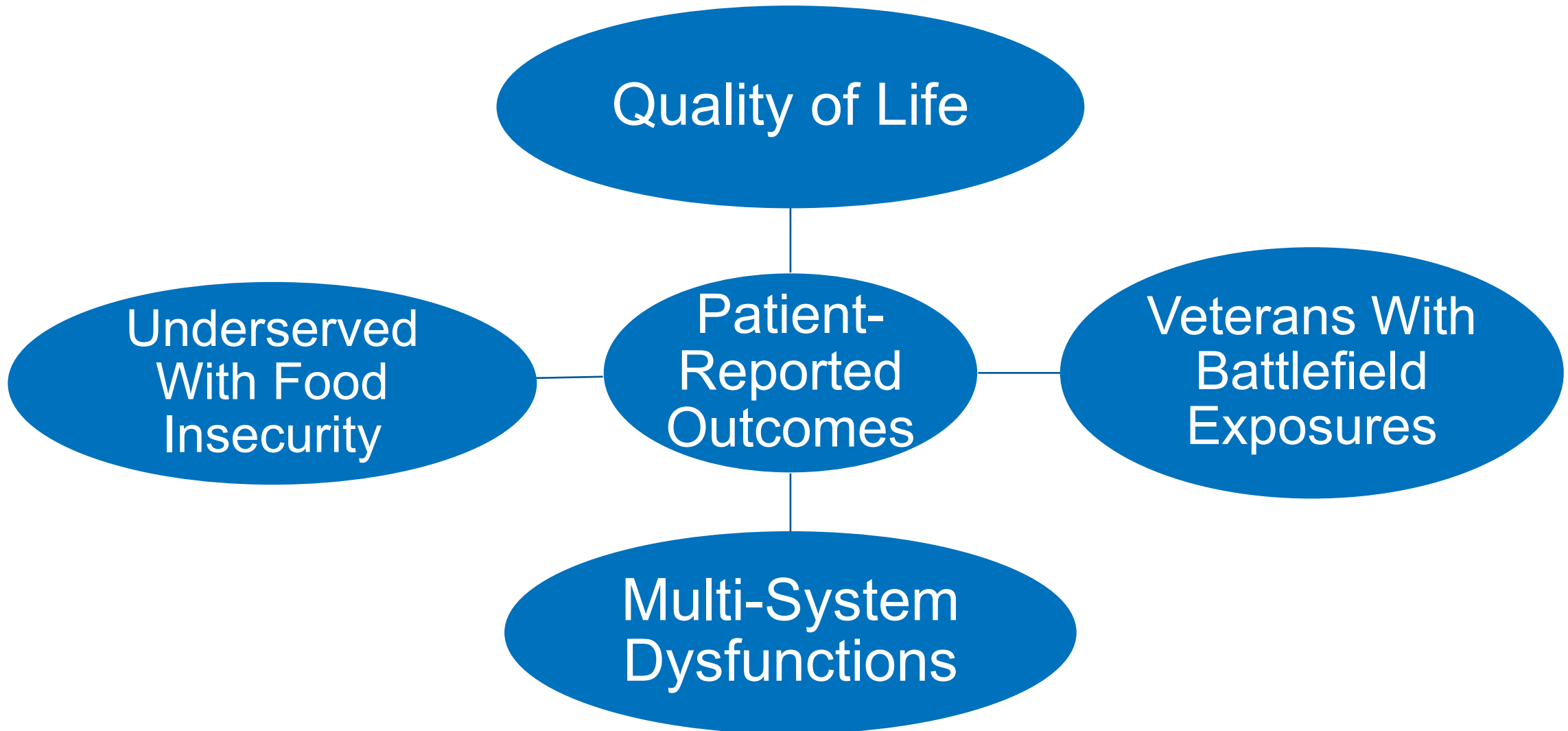
Healthcare Outcomes

Published Evidence for the Functional Medicine Model



Healthcare Outcomes

Published Evidence for the Functional Medicine Model



Clinical Focus

- The “user experience” of functional medicine.



What Patients, Clinicians, and Employers are Saying About Functional Medicine



Functional Medicine User Experience of the Clinician



Long Description: User Experience of the Clinician

This image depicts 3 circles, all slightly overlapping like a Venn Diagram. Each circle contains one of the following quotes:

So many of my patients are getting better.

I am inspired and more engaged with my patients.

This is why I went into medicine.

Functional Medicine User Experience of the Patient



Long Description: User Experience of the Patient

This image depicts 3 circles, all slightly overlapping like a Venn Diagram. Each circle contains one of the following quotes:

My Clinician communicates and partners with me.

I actively participate in my care.

The treatments are personalized for my situation.

Functional Medicine User Experience of the Employer



Long Description: User Experience of the Employer

This image depicts 3 circles, all slightly overlapping like a Venn Diagram. Each circle contains one of the following quotes:

We need our clinicians to use more nutrition and lifestyle.

We need improved patient and clinician satisfaction.

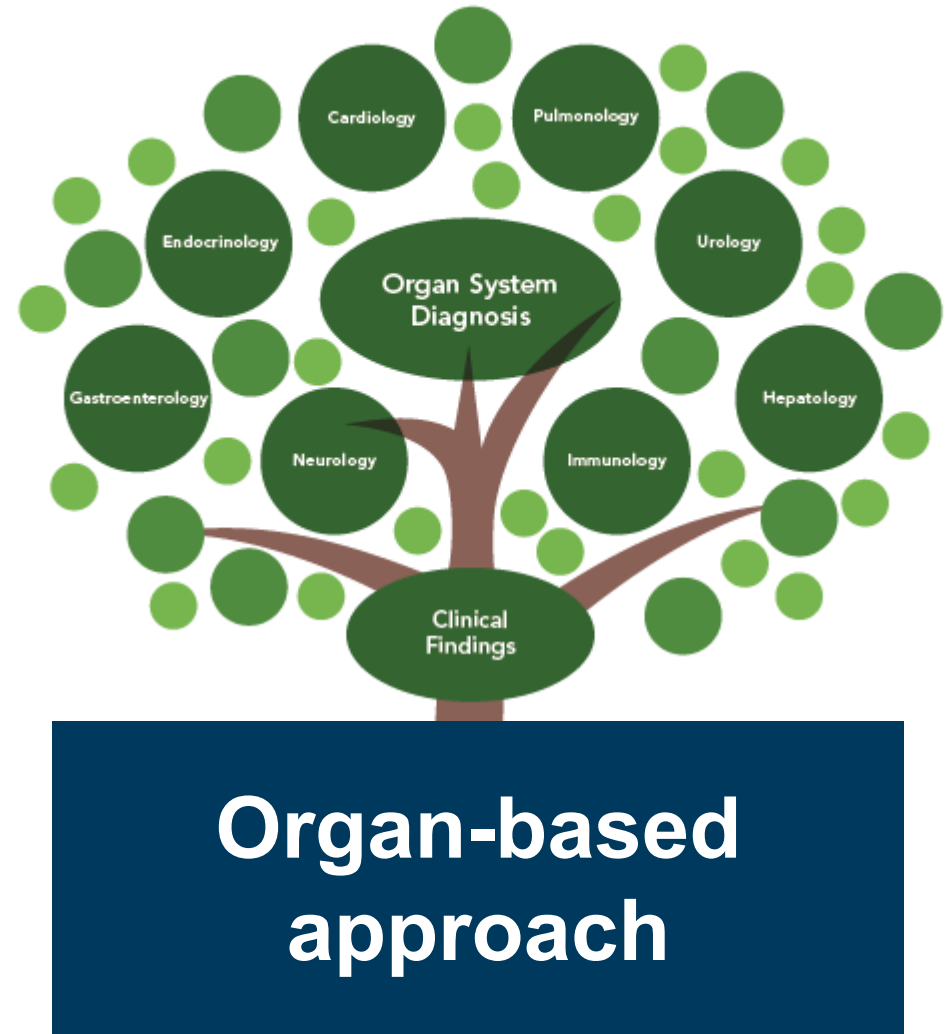
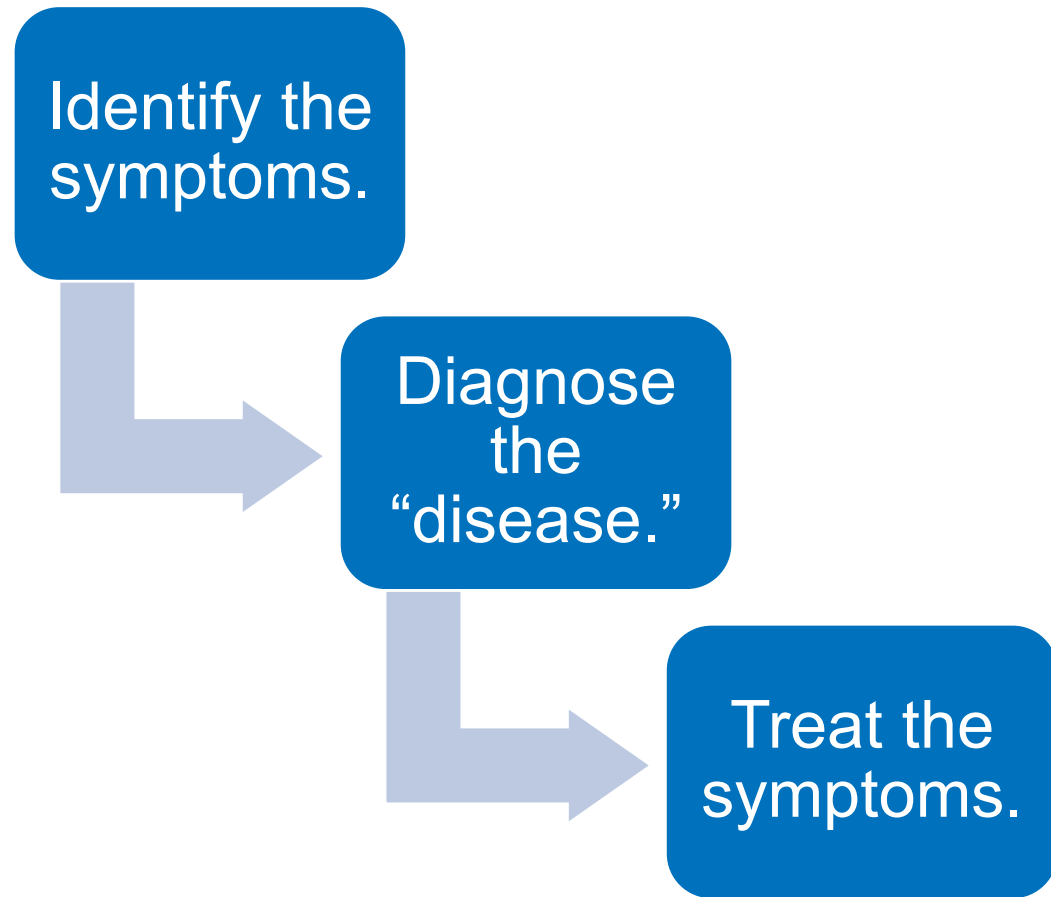
We need our patients to be more engaged in their care.

Clinical Focus

- Functional medicine is the medicine for the 21st century.



20th Century Medicine

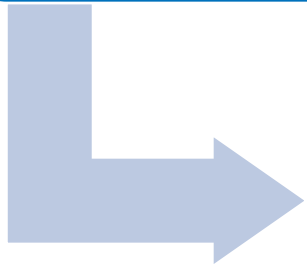


Long Description: 20th Century Medicine

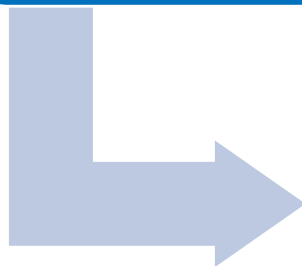
- 20th Century medicine is an organ-based approach that involves identifying the symptoms, diagnosing the disease, and treating the symptoms. If 20th Century medicine were depicted as a treat, the organ-based approach is the root system, the clinical findings are the trunk, and the organ system diagnosis including neurology, gastroenterology, endocrinology, cardiology, pulmonology, urology, hepatology, and immunology represent the branches and leaves.

21st Century Medicine

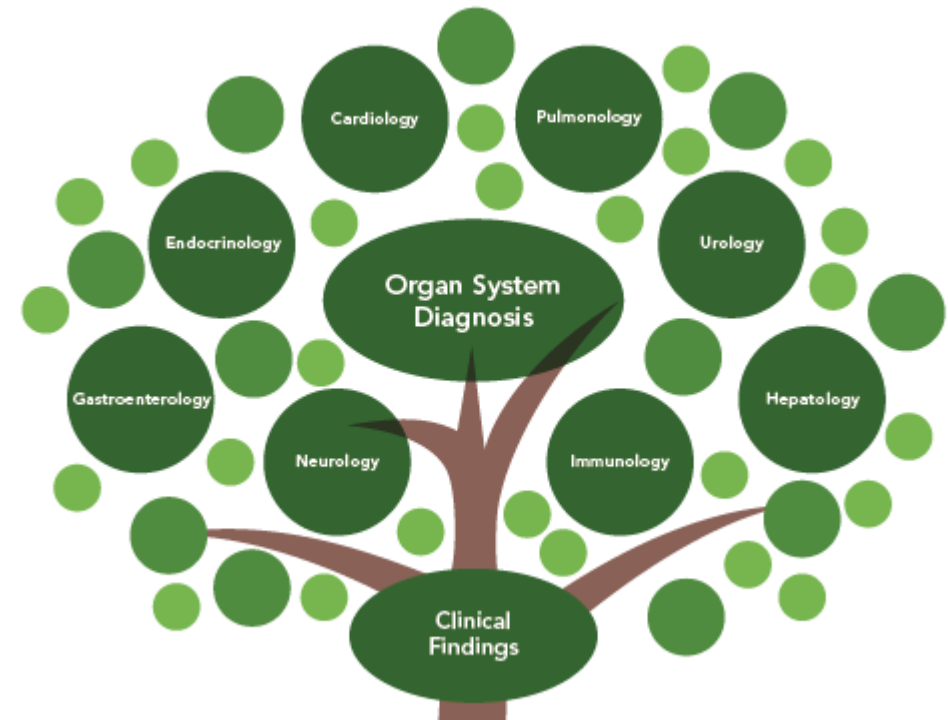
Identify the personalized root causes of the symptoms and effects.



Diagnose the dysfunctions.



Address the personalized root causes with precision.



**Systems-based
approach**

Root cause medicine

Long Description: 21st Century Medicine

- Twenty-first Century Medicine is a systems-based approach which identifies the personalized root causes of the symptoms and effects, diagnoses the dysfunctions, and addresses the personalized root causes with precision. To identify personalized root causes, the clinician must gather the patient's health events throughout their life story.

The Fundamental Organizing Systems and Core Clinical Imbalances

Assimilation

Digestion, Absorption, Microbiota/GI, Respiration

Defense & Repair

Immune system, Inflammatory processes, Infection and microbiota

Energy

Energy regulation, Mitochondrial function

Mitigation & Clearance

Toxin load, Detoxification

Communication

Endocrine, Neurotransmitters, Immune messengers

Transport

Cardiovascular, Lymphatic systems

Structural Integrity

From the subcellular membranes to the musculoskeletal system

Antecedents, Triggers, and Mediators

Genetics <> Experiences <> Exposures <> Mental <> Emotional <> Spiritual

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

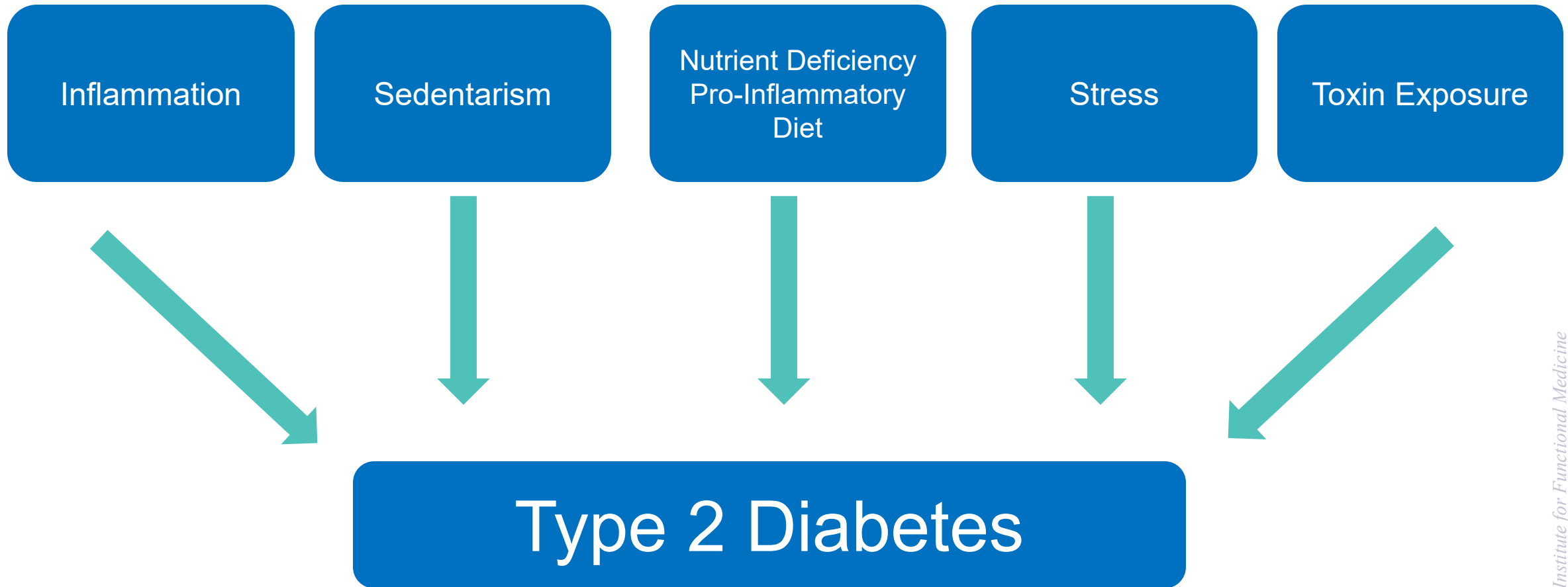
Stressors

Relationships

Long Description: The Fundamental Organizing Systems

- Toolkit item – the Functional Medicine Tree. Here the emphasis is on the fundamental organizing systems and core clinical imbalances, antecedents, triggers, and mediators, and modifiable lifestyle factors.

One Condition: Many Root Causes

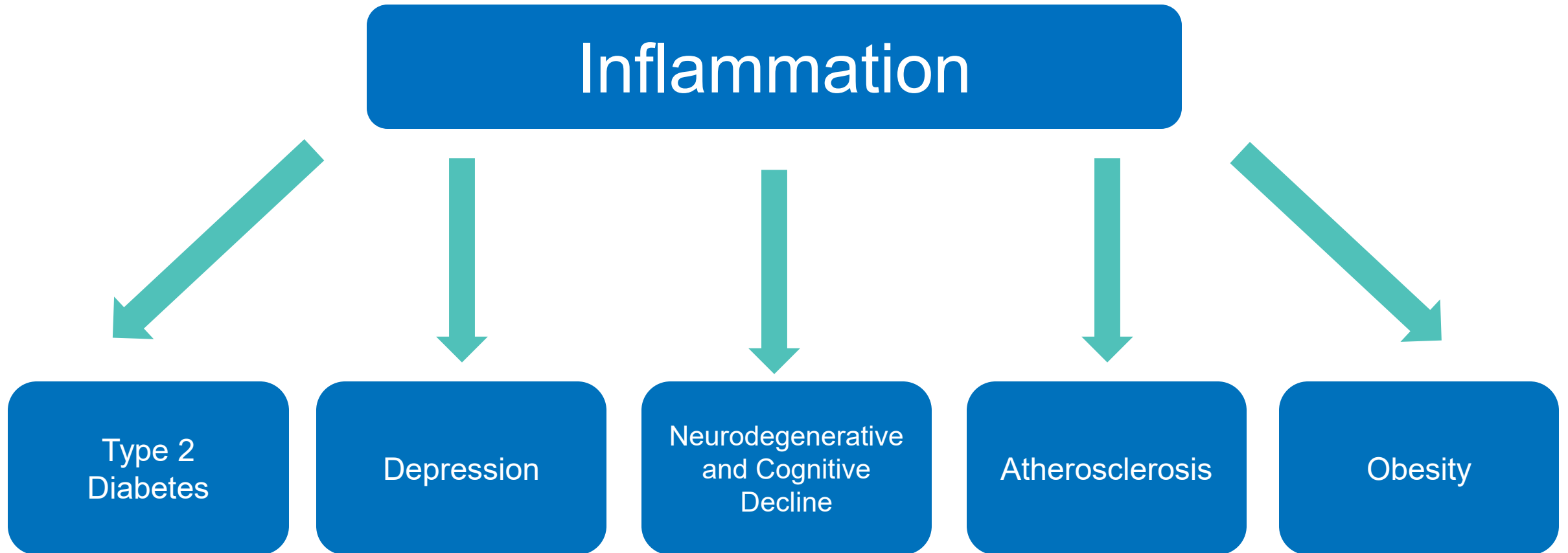


Long Description:

One Condition: Many Root Causes

- This diagram depicts how one condition such as Type 2 Diabetes, can have many root causes including inflammation, sedentarism, nutrient deficiency pro-inflammatory diet, stress, and toxin exposure.

One Dysfunction, Many Conditions



Long Description:

One Dysfunction, Many Conditions

- This diagram depicts how one dysfunction such as inflammation, may cause many conditions including Type 2 Diabetes, depression, neurodegenerative and cognitive decline, atherosclerosis, and obesity.

Functional Medicine Trains Practitioners to Become “Patient-ologists”

The patient is your specialty.

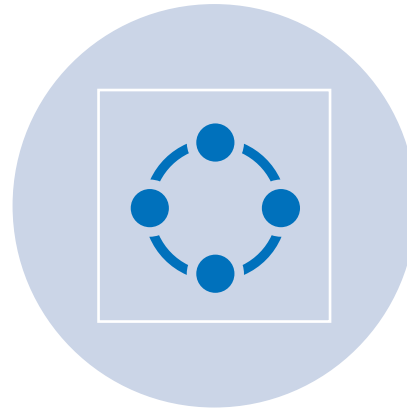


The Future of Medicine Is **Functional Medicine**

Remove what causes dysfunction.
Provide what restores function.



WHOLE-PERSON CARE



**SYSTEMS-BASED
PRECISION**



**PERSONALIZED,
ROOT-CAUSE MEDICINE**